IEEE - 54827

Tamil Speech Recognition Using XLSR Wav2Vec2.0 & CTC Algorithm

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2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT) | 978-1-6654-5262-5/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICCCNT54827.2022.9984422

Abstract- **Automatic Speech Recognition is a promising research topic with lots of real-world applications like virtual assistants, aids for physically challenged etc. Tamil language speech recognition could be potentially challenging due to the fact that there are many possible dialects, slangs and accents. This paper proposes an ASR system based on cross-lingual transfer learning in combination with CTC algorithm. The pretrained model from Facebook AI viz. XLSR Wav2Vec2.0 is used. The dataset used in this work is Common Voice Tamil, which is a crowd-sourced dataset provided by Mozilla. Our system achieves a Word Error Rate of 0.58 and Character Error Rate of 0.11.**

Keywords- **Speech recognition, ASR, cross-lingual transfer learning, XLSR Wav2Vec2.0, Common Voice Tamil**

I. INTRODUCTION

Tamil is one of the oldest languages of India belonging to the Dravidian family. It is the official language in the state of Tamil Nadu in India, Sri Lanka, Malaysia, and Singapore, with more than 74 million speakers worldwide [1]. There are many dialects of the language including Chennai Tamil, Kovai Tamil, Kumari Tamil, etc. Each dialect has its own set of vocabulary. A high level of diglossia situation exists among the speaker communities as there are always various versions of the language, namely literary and spoken. Both of them sound different and are hard to interpret without proper knowledge of the vocabulary and sentence structure. The basic structure of a Tamil sentence is subject-object-verb. Tamil being an intonation language, uses features like pitch, loudness, duration, pause, tempo, and rhythm to form the expressive meaning of the speech [1]. The rhythm of Tamil speech is syllable-based. Owing to its complex structure, semantics, and syntax, Tamil language is challenging to understand and interpret. Additionally, the language consists of multiple alphabets that represent the same alphabet otherwise, each of which is pronounced and used differently.

Speech recognition also known as Automatic Speech Recognition (ASR) is a method to convert a speech sample to a written transcript. It is a heavily used tool nowadays for many applications including mobile phones, emotion recognition, customer services, and also for people who are suffering hearing loss. Speech recognition forms a very useful counterpart in applications involving emotional speech generation [2]. It can also help physically challenged people to be able to write and type. ASR becomes one of the most important areas of research as it has vast real-life applications. There has been a lot of advancements in ASR systems in the past decade, the result of which surpasses human perception. However, majority of the available models require a lot of data and language with a massive resource for the model to learn for such good performance. The challenges associated with regional languages range from fluency, dialects, slangs, etc.

In this paper an approach is proposed for speech recognition in Tamil. The work discusses an approach with XLSR - Wav2Vec2 model and fine-tuning it for continuous speech recognition in Tamil using transfer learning. The major contributions of the paper include:

- Utilizing the idea of using cross-lingual transfer learning and making the pre-trained model XLSR Wav2Vec2.0 to learn the raw speech representations of Tamil speech audio.
- A transformer-based encoder-decoder architecture is adopted with CPC to function as the language model.
- CTC algorithm is utilized to bring out the unknown alignment mapping of the output text sequences to form meaningful sentence.

The outline of the paper is as follows: Section II. reviews the related work in Tamil speech recognition. Section III. goes about the background details of XLSR Wav2Vec2.0, CTC algorithm and the dataset used. Section IV. covers the data preparation process, and describes the proposed system design and modelling. The obtained results are discussed in Section V. Finally, Section VI. presents our conclusion and future scope.

II. LITERATURE SURVEY

Many methods bring different approaches to use textual data for speech recognition model. There is deep fusion [1] and cold fusion [2], [3] which uses pre-trained language model into a seq2seq model. The seq2seq model is initialized and requires loads of labelled data for the model training. Knowledge distillation [4], which requires end-to-end ASR models, would work far more efficiently when pretrained and finetuned models are used instead. The model in [5] builds a seq2seq model with a pre-trained encoder and multilingual decoder which is shown to perform well in ASR applications. The model proposed in [6] leverages the upside of the decoder to improve the greedy CTC outputs from the encoder where low-confidence tokens are masked. The decoder is expected to predict the tokens corresponding to those mask tokens. Decoder accomplishes the task by taking unmasked context and the acoustic representation into consideration. Hence, the idea of utilizing CTC with the encoder-decoder architecture is valid in ASR applications also. A transformer-based speech recognition model proposed in [7] predicts masked tokens on the basis of partial results iteratively. Pretraining the decoders is a challenge as they heavily depend on acoustic representation. Another aspect which has gained a lot of attention in recent years is Wav2Vec2 [8]. Cross lingual transfer that refers to use data for transfer learning by using models available for a given language with plenty of learning data to solve tasks in another language [9]. Self-supervised training approach in which models do not simply fit the labelled data but tries to interpret the input speech structure, are more suitable for domain adaptation and language transfer. In such cases contrastive predictive coding (CPC) is used. It is contrastive learning which learns to differentiate negative samples from true sample.

Summarizing, the problem with the existing models is that they fail to perform under the constraint of minimum data availability and dialect variations. Although Tamil language is one of the oldest known languages, ASR systems are still challenged by the language's complex structure [1]. Adapting an approach based on seq2seq architecture having a pre-trained encoder and multilingual decoder is far more efficient. The idea is to interpret the speech representations with already learned context and speech structure. This can be achieved through cross-lingual transfer learning across different languages and domains. In our work, we adapt a similar approach to perform speech recognition in Tamil language.

III. BACKGROUND

A. XLSR Wav2Vec2.0

Wav2Vec2.0 was the first model to demonstrate that learning the representations from the speech audio and fine-tuning on transcribed speech can achieve a good performance for speech recognition even with limited amounts of data [8]. This model basically reads raw speech input using multiple CNNs and maps the raw speech to a latent speech representation. The original model Wav2Vec2.0 was contributed by Baevski et.al. [8] which demonstrated a speech processing technique that learns speech representations directly from audio speech followed by fine-tuning of transcribed speech. Facebook AI further pretrained this model on thousands of hours of speech data covering multiple languages and released their version of the model named XLSR Wav2Vec2.0 [9]. Fig. 1 summarizes the architecture of the cross-lingual Wav2Vec2.0 model. The latent space of the model is multilingual as the model was originally pretrained on thousands of hours of speech data. Speech being continuous in nature, its latent representation is also continuous, hence it needs to be discretized. Each continuous latent vector is mapped to one of the quantized vectors 'q' from a codebook based on similarity, known as the quantized representation. A transformer-based language model is used, which maps the quantized vectors to the context representations. Some of the q vectors are actually masked and the model is expected to predict the corresponding values, based on a contrastive loss function.

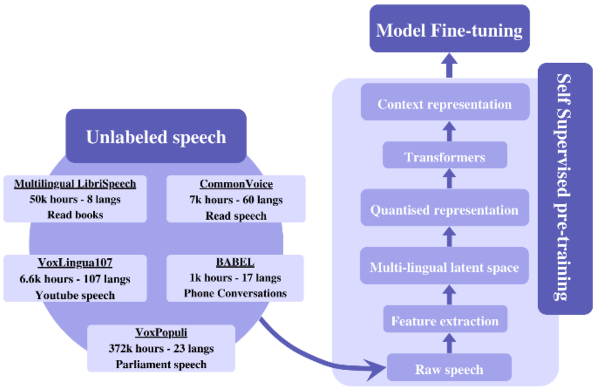

Fig. 1: XLSR Wav2Vec2.0 architecture

The raw speech representations are captured and mapped to a latent representation by a 1D convolution-based encoder followed by a GELU activation. The continuous latent representations are discretized to a finite set of speech representations via quantization. There are multiple codebooks of q vectors present from which each latent vector is mapped to form a discrete quantized vector by concatenation. Gumbel softmax function is used to enable the choosing from the codebook entries in a differentiable way. These quantized vectors are then fed to a transformer architecture to get the context information based on relative positional embedding. Relative positional embeddings are used when attention weights and the values between word i and j in the input sequence are being computed.

B. Connectionist Temporal Classification (CTC) algorithm

The CTC algorithm is a subset of neural networks that is capable of handling sequence applications such as speech recognition, handwriting recognition etc. [6]. Generally, in such problems there is a lack of knowledge about the alignment between the input (image or audio) and output sequence of characters. CTC can be used to get around such challenges even without knowing the alignment between the input and the output, hence making the training process straightforward. The aim of the algorithm is to find a mapping between the input and the output sequence. For a given input, it gives us an output probability distribution over all possible alignments of the output sequences. This distribution can be used to infer a likely output or to assess the probability of a given output. The model aims at maximizing the probability it assigns to the right answer. The CTC loss is given by equation (1) [11].

$$loss = -log\big(p(y|X)\big) = log\left(\frac{1}{p(y|X)}\right) \qquad (1)$$

where,

$p(y|X)$ is the probability distribution of all possible alignments given an input y.

Note-

- If $p(y|X)$ is approaching 0, the value of loss is approaching infinity.
- If $p(y|X)$ is approaching 1, the value of loss is approaching 0.

- Ideally, the model should be trained to optimize $p(y|X)$ value, so when the loss approaches 0, the model weight updating is stopped.

C. *Dataset*

Mozilla Common Voice – Tamil is the dataset used in this project. It contains about 7GB data with over 2 lakh data samples [12]. Though the dataset contains multiple splits of data, we make use of only the train and validation splits. The CSV file contains multiple columns of metadata, from which we utilize the transcript data and the information linking the audio files and transcripts. The audio recordings are in mp3 format and each of 4-7 seconds duration. All audio clips are sampled at 48KHz.

IV. METHODOLOGY

The section describes the steps taken to construct ASR for Tamil Language. Section A discusses the steps involved in preparing the data, section B briefs about the system design that was planned and opted and section C covers the system specifications required, parameters and hyperparameters.

A. *Data Preparation*

1. All special characters are removed from the target transcript.
2. Text in the target transcript is zero-mean unit-variance normalized.
3. XLSR-Wav2Vec2 was pretrained on Babel, Multilingual LibriSpeech (MLS)[27], and Common Voice -audio data which is sampled at 48 kHz which is later down sampled to 16kHz to feed it to the pretrained model. Common Voice data is originally sampled at 48 kHz, hence it has to be down-sampled to 16kHz for training.
4. The training samples are batch-wise padded to the longest sample.
5. The dimension of all the extracted features from the speech representations aka the feature vector size is 1.
6. Character-wise vocabulary set is formed of size 53.

B. *System Design*

The proposed system as shown in Fig. 2, learns representations from raw speech data using stacks of CNN to extract speech features. As the pretrained model Wav2Vec2.0 is used, it already has a multilingual latent space and further we are making use of it for Tamil language. The latent vectors are quantized and masked language modelling is performed. The predictions are evaluated and improved using contrastive loss. A language model with 24 transformer units and 16 attention heads, derived from Wav2vec2.0 LARGE architecture [13], is used. The context representations obtained as output is processed by the CTC algorithm. The algorithm computes probabilities for each possible alignments of the words and sentences and assigns the highest probability to the best transcript.

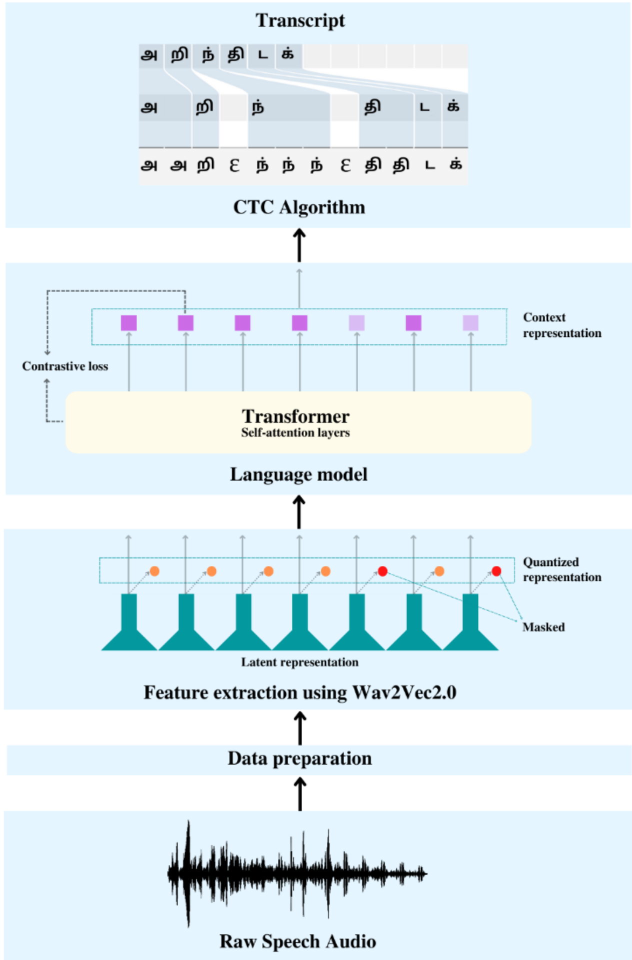

Fig. 2: Working diagram of the proposed model

C. *Modelling*

TABLE I. PROPOSED MODEL HYPERPARAMETERS, THE PARAMETERS FINE-TUNED FOR TAMIL AUTOMATIC SPEECH RECOGNITION.

Parameters	Value
Epochs	30
Learning Rate	0.0001
Train batch size	16
Hidden dropout	0.2
Evaluation strategy	Steps
Gradient accumulation	2
Gradient checkpointing	TRUE
Save steps	500
Evaluate steps	500
Logging steps	500

Pretrained XLSR Wav2Vec2.0 model is imported from Hugging Face repository [14]. Google Colaboratory is used for training our model. Our proposed system needs GPU resources to execute efficiently, hence we opt for GPU in the hardware accelerator of ColabPro engine. The specifications of the computational resources used are K80, P100, T4 GPUs with 2 virtual CPUs and 24 GB RAM. Model checkpoints are created at regular intervals of 500 samples. The model hyperparameters are as given in TABLE I.

Number of epochs considered is relatively a smaller number as the aim of the paper is to present a good result even with less training. Train batch size is selected considering the amount of GPU memory available and to avoid out of memory error.

V. RESULTS & DISCUSSION

Common Voice Tamil (test) data is used to validate the model performance. The training of the model is performed by using 80% of the dataset (Common Voice Train) for training and remaining 20% for testing. The ASR model parameters are updated and saved at every 500 steps. Word error rate (WER) and Character Error Rate (CER) are the metrics used to evaluate the model performance. Calculation of WER and CER is given by equation (2) and equation (3) respectively.

$$Word\ Error\ Rate = \frac{(S+D+I)}{N} \quad (2)$$

$$Character\ Error\ Rate = \frac{(S+D+I)}{(S+D+C)} \quad (3)$$

where,

S is the number of substitutions,
D is the number of deletions,
I is the number of insertions,
N is the total number of words
C is the total number of correct characters.

The model performance is compared with Google Speech API and the best performing model in [1]. The Google Speech API gives 55% WER as its average performance after processing all audio files of Common Voice(test) data, while our model gives shows a comparable result of 58% WER. Our proposed model gives an average CER of 11.3% which is slightly better than 11.6% CER of model proposed in [1].

The Fig. 3 gives the clarity that model is learning pretty well. Starting with the high loss in both training and validation it drops down appreciably.

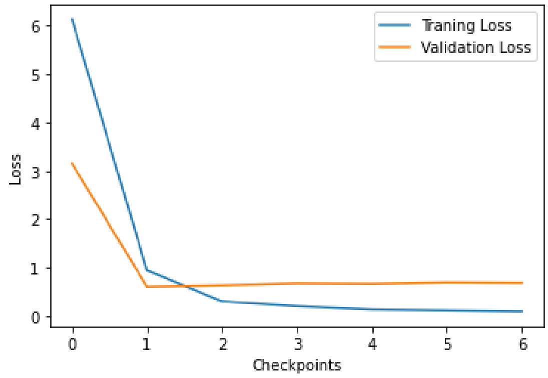

Fig. 3: Plots of training and testing losses over model checkpoints. Each model checkpoint consists of 500 samples.

TABLE II. WER AND CER COMPUTED OVER TRAINING SAMPLES IN 30 EPOCHS. THE METRICS ARE COMPUTED & UPDATED AT EVERY 500 STEPS.

Steps	WER	CER
500	0677	0.140
1000	0.633	0.130
1500	0.620	0.129
2000	0.631	0.128
2500	0.619	0.124
3000	0.595	0.119
3500	0.595	0.120
4000	0.582	0.116
4500	0.581	0.114

TABLE III. COMPARISON OF PROPOSED MODEL PERFORMANCE WITH MODEL PROPOSED IN [1] AND GOOGLE SPEECH-TO-TEXT API

Models	WER	CER
DeepSpeech(v0.7.4) LM1 [1]	0.247	0.116
Google API	0.55	-
Our proposed model	0.58	0.113

TABLE IV. EXAMPLES OF GENERATED TRANSCRIPTS BY THE PROPOSED ASR MODEL ON DIFFERENT TEST INSTANCES

Transcripts	WER	CER
Prediction: அன்று நீர் சொன்னபடி அவ்விரண்டு மூழிகையைக் **Actual:** அன்றுநீர் சொன்னபடி அவ்விரண்டு மூலிகையைச்	0.6	0.11
Prediction: என்பகைவன் உன்னாசெய் என்றன் ஆசை **Actual:** என்பகைவன் உன்னாசை என்றன் ஆசை	0.25	0.09
Prediction: தொங்கும் நரம்பின் தூழே இதைக்கே் **Actual:** தொங்கும் நரம்பின் தூளே இதைக்கேள்	0.5	0.13
Prediction: சிவனென்றும் அறியென்றும் சித்தாத்த நென்றும் **Actual:** சிவனென்றும் அரியென்றும் சித்தார்த்த னென்றும்	0.75	0.10
Prediction: முற்றும் அறிந்திடக் கூடுமென்றே அவள் **Actual:** முற்றும் அறிந்திடக் கூடுமென்றே அவள்	0.00	0.00
Prediction: நாடி வந்து நடுவிர் படுத்தாள் **Actual:** நாடி வந்து நடுவிற் படுத்தாள்.	0.25	0.05
Prediction: பேசஉண் ணாதவர் உ்புரினில் துஷ்டர் **Actual:** பேசவொண் ணாதவர் ஊரினில் துஷ்டர்	0.5	0.23
Prediction: அடமெடுத்தாடும் தமிழர் பங்கமெல்லாம் போமே **Actual:** படமெடுத்தாடும் தமிழர் பங்கமெலாம் போமே	0.5	0.07
Prediction: முட்டைபோல் உதடுகள் முன்னே தொன்ற **Actual:** முட்டைபோல் உதடுகள் முன்னே தோன்ற	0.25	0.04

TABLE V. PERFORMANCE METRICS OF THE MODEL DURING THE TRANSCRIPT GENERATION

Case	WER	CER
Best	0.00	0.00
Average	0.5	0.09
Worst	0.75	0.23

The model performance is evaluated over 4500 steps and respective WER and CER values for each checkpoint is given in TABLE II. From TABLE III. it can be infered that the obtained WER of our model is comparable to Google's

TTS API. The CER is however as good as the best CER results of DeepSpeech model in [1].

TABLE V. contains the best, average and worst WER and CER values computed on the test set. A few key observations can be made from the obtained transcripts in TABLE IV. Since each sentence in the transcript is generally 4 to 5 words long, even a single character difference brings a huge change in the WER. Hence the same situation when analysed using CER gives us a better idea about the actual differences between the model prediction and original transcript. In some cases, they are very low which means the predicted transcripts are very accurate. Tamil language consists of many similar sounding alphabets which differ in usage and hence changes the meaning of the sentence. For example, அரியென்றும் is transcribed as அறியென்றும் where "றி" and "ரி" both sounds as "ri" but the difference arises in the usage- "ரி" is used for the simple "ri" sound whereas as the "றி" is used for "tri" where the "t" is silent. The model sometimes mis-spells words because of very close pronunciation for example- செய் (pronounced as "say") is very similar sounding to சை (pronounced as "sai"). Another example can be பேசஉண் (pronounced as "pe" "sa" "un") and பேசவொண் (pronounced as "pe" "sa" "on"). The model also experiences challenges in incorporating speaking gaps, which is quite natural as human speech tends to vary in speed and rhythm, which can sometimes result in splitting of single word. For eg. அன்றுநீர் is predicted as அன்று நீர் by the model.

VI. CONCLUSION & FUTURE SCOPE

The ASR system developed by fine tuning of pretrained Wav2Vec2.0 XLSR model layered with CTC is able to recognize the speech samples and produce transcripts for the same with an average WER of 0.58 and CER of 0.11. In contrast to traditional ASR approach which require huge amounts of data and computation, we produce a simpler and computationally less expensive Tamil ASR. Use of a pre-trained model eases out the complexity of training. The major challenge encountered in Tamil speech recognition is to incorporate the phonetic understanding of the language. Our model flusters slightly when it comes to choosing the correct alphabets to form the transcripts as there are many similar sounding alphabets in Tamil. Another challenge faced by our model is word-splitting because human speech varies in rhythm and speed.

In the future there is scope for the creation of a better dataset by incorporating the specifics of Tamil language keeping the pronunciation and spellings in focus. The model can be extended to learn to resolve the pronunciation conflicts of the words and generate more accurately spelt transcripts. Our model can also be extended to be a full-fledged application performing real time transcriptions. It can be used to potentially aid visually challenged people in many areas including work space typing jobs for visually impaired people, ASR can be extended so that it can edit document with speech commands.

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